

HARIS WAHEED BHATTI

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Full-time work rights as a student until 30/09/2026, transitioning to an EU job-seeker permit thereafter; EU Blue Card eligible.

PROFILE

Civil engineer with a European Master's in Building Information Modelling, building software for the AEC industry - IFC schemas, graph databases, and LLM agent systems. Author of a published open-source package for IFC-to-Neo4j conversion.

PROJECTS

- ifctoneo4j** - Python package (PyPI) | *Python, IfcOpenShell, Neo4j, Cypher* 2026
- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into **Neo4j** property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published and installable via `pip install ifctoneo4j`
- BIMGraphExplorer** - LLM agent for IFC question answering | *Python, MCP, Neo4j* 2026
- Adaptive querying system exposing four read-only tools over the Model Context Protocol; the model profiles the graph schema and issues iterative Cypher queries conditioned on prior results
 - Raised answer accuracy by 33 percentage points and source-transparency from 12.5 % to 99.7 % on a fixed model and dataset through prompt design alone
 - Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge, same scope; 23 IFC datasets, 367 questions, 15 projects), achieving transparency at parity (99.7 % versus 100 %) and outperforming the baseline on 5 of 23 datasets at 72.0 % versus 82.3 % overall accuracy
- Blockchain-Based BIM Document Management** | *Solidity, Ethereum (Arbitrum Sepolia)* 2024
- Smart contract system to digitally sign and manage construction documents, with off-chain storage linked on-chain for immutability, designed to streamline dispute resolution under **FIDIC** contracts; funded by a competitive PKR 100,000 grant from the Pakistan Engineering Council
- Federated BIM Model** - Pakistani-Themed Expo, Guimarães | *IFC, Autodesk Construction Cloud* 10/2025
- Led BIM coordination across architecture and MEP disciplines; delivered a federated model with clash detection and compliance against Employer's Information Requirements (EIR) under **ISO 19650**

EDUCATION

- European Master in Building Information Modelling (BIM A+)** 10/2025 - 07/2026
University of Ljubljana, UL FGG *Slovenia*
- Field of study: Information Management in Construction; completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade: 9.27/10
 - Thesis:** *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. [Repository Link](#)
 - Self-funded living costs through part-time employment throughout the programme
- BE Civil Engineering** 09/2020 - 06/2024
National University of Sciences and Technology (NUST) *Pakistan*
- CGPA: 3.25/4.00

SKILLS

Programming & Data: Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git
AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking
BIM & AEC: IFC (IFC2X3, IFC4, IFC4X3), IfcOpenShell, Linked Building Data ontologies, ISO 19650, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, web-ifc
Languages: English (fluent), Slovene (A1, actively learning)

EXPERIENCE

- Engineering Internships** 2023
Pakistan & USA (remote)
- Contract Management Intern**, Engineering and Management Services (PK): reviewed contract reports and arbitration documentation under **FIDIC** conditions; presented findings to the internal legal team
 - Civil Engineering Intern**, Fatima Fertilizer Company (PK): modelled a 6,804 m² rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement
 - Research Intern** (remote), Southern Illinois University Edwardsville (US): synthesised academic literature to support a faculty-mentored research proposal